

Sample Paper
FINALTERM EXAMINATION
Fall 2023
CS610P - Computer Networks (Practical)

Time: 60 min
Marks: 40

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Solved by Qalb-e-Momin

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Question No: 1 (Marks: 01) - Please choose the correct option

1. Which one of the following addresses is used to uniquely identify each device in a TCP/IP network?
 - a) TCP
 - b) **IP**
 - c) FTP
 - d) UDP

2. _____ is the correct IP address.
 - a) 192.168.0.256
 - b) 10.25.0.300
 - c) **192.168.0.250**
 - d) 172.256.10.2

Reason:

In IPv4, each octet can only range from 0 to 255.

3. Which of the following cable is used to connect a PC with a switch/hub and router with a switch?
 - a) **Straight-through**
 - b) Crossover
 - c) Roll over
 - d) Console

PC to PC & switch to switch

4. The maximum length for thick coaxial cable may be up to _____.
 - a) 100 meters
 - b) 600 meters
 - c) **500 meters**
 - d) 1000 meters

5. Which of the following connectors is used with twisted pair cable?
- a) RJ-11
 - b) RJ-45
 - c) RJ-55
 - d) RJ-65
6. _____ device can analyze incoming data packets.
- a) Bridge
 - b) HUB
 - c) Repeater
 - d) Amplifier
7. _____ device is used to connect two different types of computer networks.
- a) Hub
 - b) Switch
 - c) Router
 - d) Repeater

Hub → Connects devices within the same network.

Switch → Connects devices within the same network but more efficiently than a hub.

Repeater → Extends signal range, doesn't connect different networks.

8. Which of the following types of protocol determines the path of a packet through an inter-network?
- a) Routing
 - b) Rolling
 - c) Rated
 - d) Relevant
9. Suppose if you have large network with so many devices then _____ routing will be preferred.
- a) Auto
 - b) Static
 - c) Dynamic
 - d) Straight
10. _____ allows users to capture and analyze the traffic running on a computer network.
- a) Wire shark
 - b) Packet tracer
 - c) IDS
 - d) Router
11. Which of the following protocol used to translate an IP address to a hardware address?
- a) ARP

- b) DNS
- c) FTP
- d) TCP

12. IP address has _____ different classes.

- a) 3
- b) 4
- c) 5
- d) 6

13. _____ bits are used in Class C for network portion.

- a) 8
- b) 16
- c) 24
- d) 32

14. Which of the following number of bits will be used for host portion in class A?

- a) 8
- b) 16
- c) 24
- d) 32

15. Which of the following address belongs to class A?

- a) 50.20.1.0
- b) 136.25.2.10
- c) 199.3.2.1
- d) 236.12.0.1

16. Which of the following address belongs to class C?

- a) 200.40.1.2
- b) 10.25.3.0
- c) 172.16.2.1
- d) 235.2.1.0

17. Which of the following is a loop back address?

- a) 127.0.0.1
- b) 192.168.2.5
- c) 172.16.2.3
- d) 10.0.1.2

18. Which of the following is a private IP address?

- a) 192.168.2.1
- b) 224.25.1.2
- c) 172.85.2.1
- d) 89.2.5.1

19. Suppose if we borrow 3 bits from host portion in class c then _____ subnets and _____ usable hosts per subnet will be created.

- a) 4 and 62
- b) 2 and 126

- c) 8 and 30
d) 16 and 14
20. Identify the correct subnet mask for class C.
a) 255.0.0.0
b) 255.255.0.0
c) 255.255.255.0
d) 255.255.255.255
21. Which of the following is a multicast address?
a) 224.0.2.1
b) 192.168.2.1
c) 173.16.52.1
d) 10.25.4.0
22. _____ is the correct subnet mask for the network address 192.29.0.0 to get 10 subnets?
a) 255.255.255.128
b) 255.255.255.192
c) 255.255.255.240
d) 255.255.255.224
23. Which of the following is a correct CIDR address notation?
a) 192.168.1.1/24
b) 192.168.1.1\24
c) 192.168.1.1-24
d) 192.168.1.1!24
24. What is the purpose of the slash notation in CIDR addressing?
a) A) To indicate the number of IP addresses in a subnet
b) B) To indicate the network address of a subnet
c) C) To indicate the broadcast address of a subnet
d) D) To indicate the subnet mask of a subnet
25. Each value in IPV6 address is separated by _____ sign.
a) Dolor sign
b) Colon
c) Percentage
d) Asterak

Question No: 21 (Marks: 05)

Suppose a currency exchange company wants to design a computer network. The company has its head office at Lahore and recently opened a branch in Faisalabad. The head office comprising of 3 floors that will be connected with Lahore branch. Employees in Lahore are in need to access the confidential information from the server located in Lahore. By keeping in mind the scenario You are required to answer the following questions:

In Lahore branch, which type of intelligent networking device will be used to connect computers with each other?

1. Name the address that will be used for communication among the devices in LAN of Lahore.
2. Which secure connection will be created for communication between Lahore and head office computers?
3. Name the networking device that will be used in head office to connect all three LANs with each other.
4. Which networking device will be used for creating connection between Lahore and Faisalabad office?

Solution:

1. In Lahore branch, which type of intelligent networking device will be used to connect computers with each other?

Switch (an intelligent device that connects computers in a LAN and forwards data based on MAC addresses).

2. Name the address that will be used for communication among the devices in LAN of Lahore.

MAC address (Media Access Control address).

3. Which secure connection will be created for communication between Lahore and head office computers?

VPN (Virtual Private Network) — provides encrypted, secure communication over public or private networks.

4. Name the networking device that will be used in head office to connect all three LANs with each other.

Router (connects different networks, including multiple LANs).

5. Which networking device will be used for creating connection between Lahore and Faisalabad office?

Router (used for interconnecting remote offices over WAN links).

// sara likhny ki zarorat nhi hy... just write the highlited names

The company ABC has its main office in Karachi and two smaller branches in different locations. Each branch has its own LAN with 100 computers. The main office has 300 computers connected to its LAN. The LANs of each branch are connected to the head office using Wide Area Network (WAN), which allows the branches to communicate and share resources with the head office. Furthermore, to ensure the connectivity of devices in all LANs, the network administrator regularly performs tests. After reading scenario you are required to answer the following questions:

1. Which Type of address will be used for communication among devices in a LAN.
2. Which type of address will be used for communication between different branches.
3. Which type of command will be used to ensure the connectivity?

Solution:

1. Which type of address will be used for communication among devices in a LAN?

MAC Address (Media Access Control address).

2. Which type of address will be used for communication between different branches?

IP Address (Internet Protocol address).

3. Which type of command will be used to ensure the connectivity?

ping command (used to test network connectivity between devices).

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Question No: 23 (Marks: 05)

How many minimum IP addresses will be purchased by the ISP to accommodate 450 customers assuming that ISP can only purchase IP addresses range as a block of 2^n . Suppose in future, ISP wants to extend its service in small colony to accommodate more customers, then how many customers will be accommodated with the remaining available addresses after accommodating 450 customers?

Solution:

- ❖ **Minimum IPs to purchase: 512**
- ❖ **Extra customers possible later: 62**

HOW?

Step 1: Find the smallest block size (2^n) that can hold 450 customers

- Block sizes: $2^8 = 256 \rightarrow$ too small.
- $2^9 = 512 \rightarrow$ enough to hold 450.

So, minimum IP addresses to purchase = 512. ✓

Step 2: Find remaining addresses after serving 450 customers

- Purchased: 512
- Used: 450
- Remaining = $512 - 450 = 62$ addresses.

Step 3: How many more customers can be accommodated?

- Each customer needs 1 IP address (assuming 1 per customer).
- Remaining 62 customers can be accommodated. ✓

Question No: 24 (Marks: 05)

Considering the following IP address and its corresponding subnet mask:

IP Address: 192.18.5.202

Subnet Mask: 255.255.255.248

You are required to provide the following information in the given table:

No. of subnet bits	
Total No. of subnets	
Total No. of useable hosts per subnet	
Network address of this subnet	

Solution:

No. of subnet bits	5 bits
Total No. of subnets	32 subnets
Total No. of useable hosts per subnet	6 host
Network address of this subnet	192.18.5.202

Question No: 25 (Marks: 05)

Suppose you have designed an office network of two floors that are attached with each other using router device in the Packet Tracer Tool. You are required to write all steps which will be used to assign IP address and default gateway to a router device.

Solution:

1. Open the Project in Packet Tracer
2. Access the Router CLI
3. Enter Global Configuration Mode
4. Assign IP Address to Router Interfaces
5. Assign Default Gateway to PCs
6. Save the Router Configuration
7. Test the Connectivity

Question No: 26 (Marks: 05)

Identify the IP address class for each IP address in the following table:

Binary value of first octet	IP address class
-----------------------------	------------------

10.25.3.56	
224.25.21.0	
198.125.100.1	
75.45.23.1	
245.56.89.20	

Solution:

Binary value of first octet	IP address class
10.25.3.56	A
224.25.21.0	D
198.125.100.1	C
75.45.23.1	A
245.56.89.20	E

IP Class Ranges Recap

✓.✓.✓.imp

- **Class A:** 1–126 (binary starts with 0)
- **Class B:** 128–191 (binary starts with 10)
- **Class C:** 192–223 (binary starts with 110)
- **Class D:** 224–239 (binary starts with 1110) → Multicast
- **Class E:** 240–255 (binary starts with 1111) → Experimental

Question No: 27 (Marks: 05)

Consider the IP address 215.35.5.1/25 and provide the following information:

Total number of subnets

Total number of useable hosts

Solution:

- **Total number of subnets = 2.**
- **Total number of usable hosts = 126 per subnet (252 usable addresses across both subnets).**

HOW?

Given: 215.35.5.1/25

Assume classful default for 215.x.x.x is **Class C (/24)**.

1. Total number of subnets

- Subnet bits = 25 - 24 = 1.
- Total subnets = $2^1 = 2$.

2. Total number of usable hosts

- Host bits = 32 - 25 = 7.
- Total addresses per subnet = $2^7 = 128$.
- Usable hosts per subnet = $128 - 2 = 126$ (subtract network and broadcast).
- If you mean *across all subnets*, total usable hosts = $126 \times 2 = 252$.

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